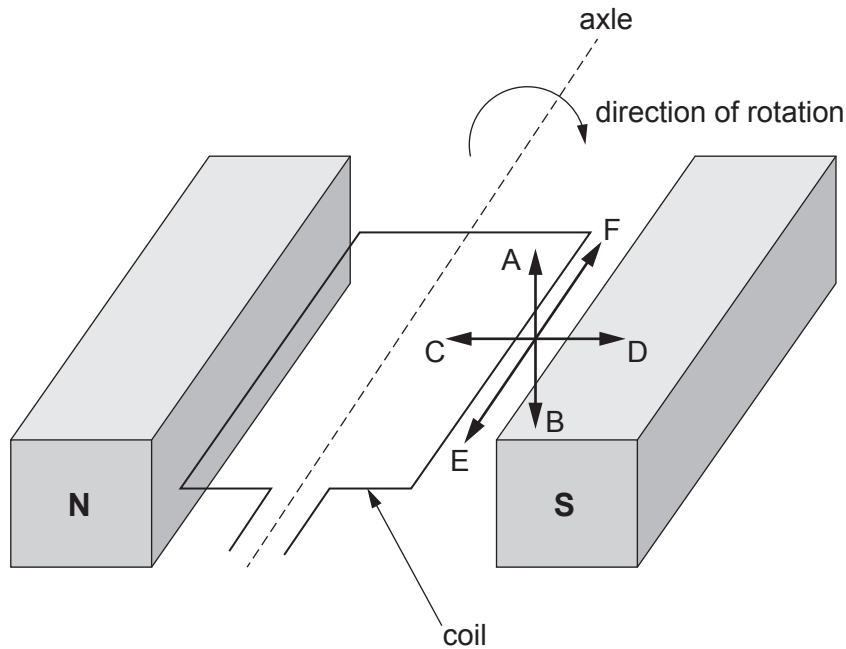


5. The operation of electric generators can be explained by electromagnetic induction.

(a) The diagram shows part of an a.c. generator.



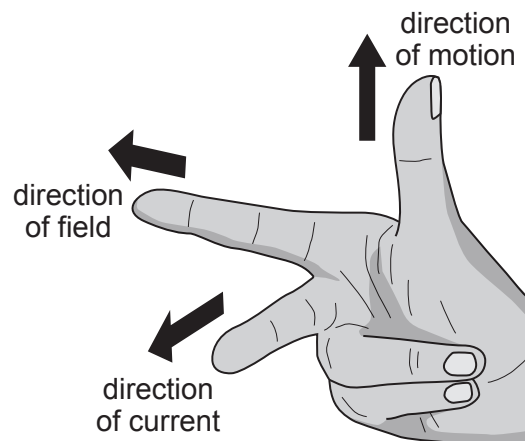
(i) State the direction (A, B, C, D, E or F) in which the magnetic field acts. [1]

Direction =

(ii) State the direction (A, B, C, D, E or F) in which the right hand side of the coil moves. [1]

Direction =

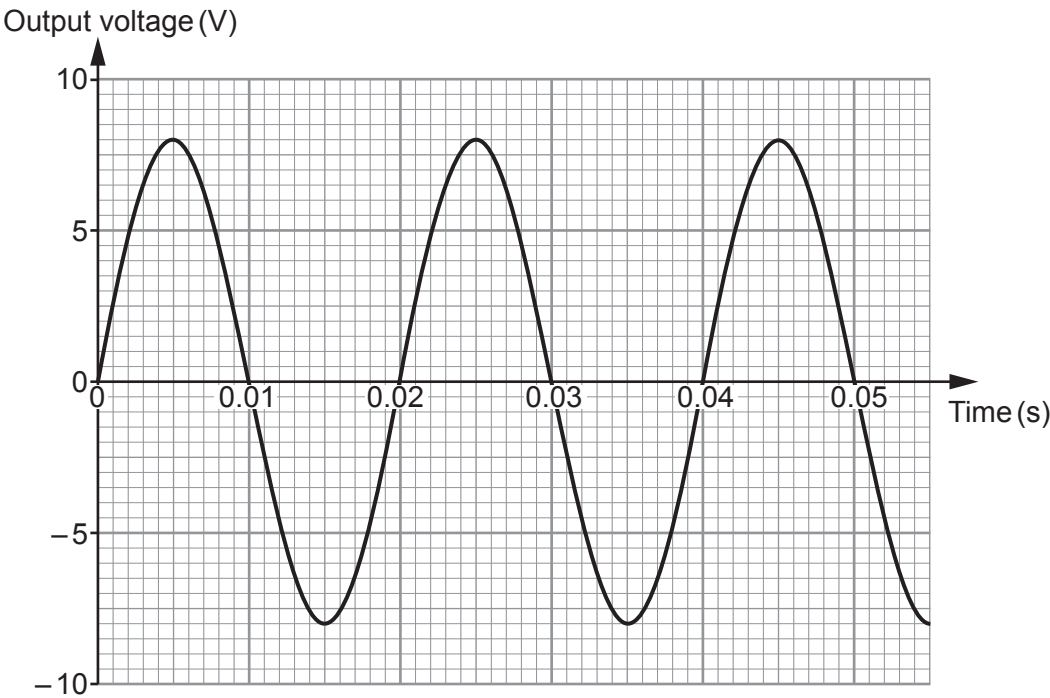
(iii) Use Fleming's right hand rule to determine the direction (A, B, C, D, E or F) of the induced current in the right hand side of the coil. [1]



Direction =



(iv) When the coil spins, a voltage is produced.
This is shown on the graph.



I. Use the graph to find the maximum voltage produced by the generator. [1]

maximum voltage = V

II. Use the graph to find the time for one rotation (spin) of the coil. [1]

time = s

III. **Complete the table** below to show the effect of each change made.
Use the words **increases**, **decreases** or **no effect**.
Three boxes have been completed for you. [3]

Change made	Effect on maximum voltage	Effect on time for 1 rotation (spin)
Stronger magnets	increases	no effect
More turns on the coil		no effect
Coil spins slower		



(b) Transformers also work by electromagnetic induction.

(i) Tick (✓) the boxes next to the **three** correct statements about transformers. [3]

Transformers change a.c. to d.c.

☐

Transformers change d.c. voltages.

☐

The core is made from iron.

☐

There is an electric current in the core.

☐

The primary coil creates an alternating magnetic field.

☐

Transformers can increase or decrease voltages.

☐

(ii) A transformer increases voltage from 12 V to 36 V.
The primary coil contains 20 turns of wire.
Use the equation:

$$N_2 = N_1 \times \frac{V_2}{V_1}$$

to calculate the number of turns in the secondary coil, N_2 . [2]

$N_2 = \dots\dots\dots$

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